

AI Text-to-Video SaaS Platform - Complete Structure

This document contains the complete professional structure for building an AI-powered Text-to-Video SaaS platform with:

- AI video generation
- Voiceovers and subtitles
- Copyright detection
- SEO automation
- YouTube and Instagram publishing
- Secure storage and analytics

Main Project Structure

```
visionforge-ai/
├── frontend/
├── backend/
├── ai-services/
├── video-engine/
├── copyright-system/
├── publishing-system/
├── seo-engine/
├── subtitle-engine/
├── thumbnail-engine/
├── storage-system/
├── deployment/
├── docs/
├── tests/
└── README.md
```

Frontend Structure

```
frontend/
├── public/
├── src/
│   ├── api/
│   ├── components/
│   ├── pages/
│   ├── hooks/
│   ├── contexts/
│   ├── layouts/
│   ├── styles/
│   ├── utils/
│   ├── App.js
│   └── main.js
├── package.json
└── vite.config.js
```

Backend Structure

```
backend/
├── auth/
├── users/
├── ai_engine/
├── video_engine/
└── voice_engine/
```

```
■■■ subtitle_engine/
■■■ seo_engine/
■■■ thumbnail_engine/
■■■ copyright_checker/
■■■ publish_system/
■■■ analytics/
■■■ database/
■■■ storage/
■■■ security/
■■■ config/
■■■ main.py
```

AI Services Structure

```
ai-services/
■
■■■ openai/
■■■ runway/
■■■ stability_ai/
■■■ gemini/
■■■ claude/
■■■ orchestrator/
```

Video Engine Structure

```
video-engine/
■
■■■ scenes/
■■■ templates/
■■■ transitions/
■■■ animations/
■■■ assets/
■■■ music/
■■■ fonts/
■■■ render_queue/
```

Copyright Detection Structure

```
copyright-system/
■
■■■ audio-fingerprint/
■■■ frame-analysis/
■■■ ai-detection/
■■■ similarity-engine/
■■■ reports/
```

SEO Engine Structure

```
seo-engine/
■
■■■ youtube/
■■■ instagram/
■■■ tiktok/
■■■ keyword-analysis/
■■■ trend-analysis/
```

Publishing System Structure

```
publishing-system/  
■  
■■■ youtube/  
■■■ instagram/  
■■■ tiktok/  
■■■ oauth/  
■■■ token_manager/  
■■■ scheduler/
```

Storage Structure

```
storage-system/  
■  
■■■ uploads/  
■■■ renders/  
■■■ thumbnails/  
■■■ subtitles/  
■■■ temp/  
■■■ backups/
```

Deployment Structure

```
deployment/  
■  
■■■ docker/  
■■■ nginx/  
■■■ kubernetes/  
■■■ ci-cd/  
■■■ monitoring/  
■■■ backups/
```

Complete User Flow

```
User Login  
↓  
Enter Prompt  
↓  
AI Script Generation  
↓  
Scene Generation  
↓  
Voiceover Creation  
↓  
Subtitle Generation  
↓  
Video Rendering  
↓  
Copyright Detection  
↓  
SEO Generation  
↓  
Thumbnail Generation  
↓  
Preview  
↓  
Download OR Publish  
↓  
Analytics Tracking
```

Final Product Vision:

A complete AI-powered SaaS platform that automates video generation, copyright checking, SEO optimization, and publishing for creators on YouTube, Instagram, Shorts, Reels, and future social media platforms.